

Project Title: *Development of Laboratory management web portal*

Problem Statement:

ICAR-IIRR is a premiere rice research Institute funded by Govt. of India. It houses state of the art research laboratories each focusing on a specific research program. Each laboratory is headed by a Principal Investigator/Lab Administrator. The Lab admin manages all resources including people, project, publications, and inventory of the laboratory for smooth execution of research experiments to achieve the objectives. Therefore, a complete suite (web portal) with a dashboard for real time monitoring of status of all resources is essential for the Lab Admin for the efficient deployment resources to increase the productivity of the lab.

Pain Points: Currently, the information about all these resources is scattered in different hard copies of files, soft copies of documents stored in different folders/computers making it tedious and time consuming for the lab Admin to manage them and for checking the status in real time.

Solution: Development of a complete suite (web portal) with a dashboard for real time monitoring of status of all resources

Approach: Designing a Laboratory management web portal with an integrated dashboard requires a centralized approach to track, and deploy laboratory sources like people, projects, and inventory etc. The system should provide high-value, actionable insights to reduce duplication of tasks, planning timely purchase of inventory, recruitment of people, etc. It should ideally have following features:

1. **The Dashboard (Command Center):** The dashboard should provide a high-level overview and real-time status of laboratory resources.
2. **A Centralized Repository** of all resources:
 - **People:** It includes all categories like Lab Admin, Scientist, Technician, Post Doc, RA, SRF, JRF, Project Assistant, PhD student, MSc Student along with date of joining and data of relieving, etc.
 - **Projects:** Project Name, Project Code, Type of Project, Funding Source, Date of Start, Date of Closing, Report Status.
 - **Publications:** Publication title, Project, Year, Journal, NAAS Score
 - **Inventory:** Permanent Equipment, Semi-Permanent, Chemicals, Glassware, Plastic ware, Miscellaneous items with following details for each item: Date of purchase, Make, Model, Cat No, Qty, Price, Total Quantity in Stock.
3. **Analytics & Reporting:** Ability to generate Custom Reports by Resource classes
4. **Technical Requirements & User Experience**
 - **Security & Compliance:** High-security encryption and Access control (e.g., HTTPS, 2FA) to protect sensitive IP data.
 - **Integration:** Integration with Microsoft Office (Word/Excel/Outlook) for filing and document creation.
5. **Implementation Steps**
 - **On boarding:**

- Creation of database files, data entry offline/online through portal's user interface with role based login credentials.
- Inventorying all lab resources, and entering in to digital databases
- **System Configuration:**
 - Setting up customized dashboards, workflows, and automated reminders.

This structure should ensure that the Lab management platform is not just a repository, but a strategic tool to protect and maximize the value of IP assets.

Data Base Structure:

Database-1: People

Fields: Lab Admin, Scientist, Technician, Post Doc, RA, SRF, JRF, Project Assistant, PhD student, MSc Student along with date of joining and data of relieving,

Database -2 – Projects

Fields: Project Name, Project Code, Type of Project, Funding Source, Date of Start, Date of Closing, Report Status.

Data Base-3 – Publications

Fields: Project, Year, Journal, NAAS Score

Data Base-4 – Inventory

Fields: Inventory Class (drop down: Permanent Equipment, Semi-Permanent, Chemicals, Glassware, Plastic ware, Miscellaneous items), Date of purchase, Make, Model, Cat No, Qty, Price, Total Quantity in Stock.